

Constraint-First MOND in Minimally Modified Gravity:

A Two-Degree-of-Freedom Hamiltonian Construction

Carl P. Zimmerman
Briar Creek Tech

<https://github.com/carlzimmerman/zimmerman-formula/>

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Research status: conditionally closed construction; not a globally closed or manifestly four-dimensional covariant theory.

Abstract

We construct and analyze a Hamiltonian modification of general relativity designed to reproduce a prescribed MOND interpolation law while retaining only the two tensorial gravitational degrees of freedom on its generic nondegenerate branch. The model is formulated as a spatially covariant, preferred-foliation theory in the framework of minimally modified gravity (MMG), rather than as a manifestly four-dimensional generally covariant local action. The central idea is to impose the MOND equation itself as an elliptic gravitational constraint instead of introducing a propagating scalar sector. The target constitutive relation is $\mu(y) = 1 - e^{-y}$, with $y = |\nabla\Psi|/a_0$ in the weak-field limit. A local primitive $\mathcal{G}(y) = y^2 + 2(1+y)e^{-y} - 2$ satisfies $\mathcal{G}'(y)/(2y) = 1 - e^{-y}$. The Hamiltonian phase space is reduced by four scalar second-class constraints in addition to the spatial-diffeomorphism constraint structure. The corresponding Dirac matrix has determinant $(L_N K)^2$, where L_N is the principal MOND lapse operator and K is the principal bracket of the inhomogeneous scalar auxiliary pair. On the generic branch $y > 0$ and $k \neq 0$, both are nondegenerate and the degree-of-freedom count is $20 - 12 - 4 = 4$ phase-space dimensions, i.e. two local gravitational degrees of freedom. The transverse-traceless sector remains the Einstein sector at quadratic order with $\omega^2 = c^2 k^2$ and positive kinetic coefficient. The weak-field limit reduces exactly to $\nabla \cdot [(1 - e^{-|\nabla\Psi|/a_0}) \nabla\Psi] = 4\pi G \rho_b$. The construction is therefore a conditionally closed two-degree-of-freedom MOND model. Open issues are the homogeneous $k = 0$ sector, the exactly degenerate $y = 0$ branch, the preferred-foliation interpretation, and the fully relativistic matter-conservation structure. These limitations are stated explicitly rather than hidden in the closure claim.

1 Introduction

A relativistic MOND construction must do more than reproduce a phenomenological interpolation function. It must specify a dynamical theory, identify its physical phase space, control unwanted degrees of freedom, and possess a well-defined weak-field limit. These requirements are in tension because many scalar or nonlocal realizations that reproduce the desired MOND response introduce additional canonical modes.

The present construction takes a constraint-first route. Instead of introducing a propagating MOND scalar, the MOND relation is imposed as an elliptic gravitational constraint in a preferred spatial foliation. The remaining inhomogeneous scalar metric pair is then removed by a second-class constraint pair. The resulting theory belongs to the Hamiltonian class of minimally modified

gravity (MMG), where spatial covariance is retained while the usual refoliation symmetry is replaced by additional auxiliary constraints. The general MMG framework with four scalar auxiliary constraints was developed by Yao, Oliosi, Gao, and Mukohyama, who showed through a Hamiltonian analysis that four scalar auxiliary constraints can yield theories with only the two tensorial gravitational degrees of freedom of general relativity. [2]

A more recent construction by Sangtawe, De Felice, and Karwan uses Laplacian auxiliary constraints to remove homogeneous-multiplier ambiguity and obtains a generic branch with no propagating gravitational scalar or vector mode. [3] The present work uses the same general constraint philosophy but supplies a MOND-specific nonlinear elliptic constraint whose weak-field constitutive law is fixed exactly.

The result should be regarded as a mathematical field-theory construction with a checked generic branch, not as a completed replacement for general relativity. The distinction is central to the paper.

2 Why a constraint-first construction

An earlier causal nonlocal route considered a retarded response field

$$\Phi = \square_{\text{ret}}^{-1} (R_{\mu\nu} U^\mu U^\nu), \quad (1)$$

with a constitutive invariant built from $\nabla_\mu \Phi$. Its weak-field sector generated the desired interpolation function, but unrestricted localization introduced an indefinite auxiliary kinetic matrix. Although the causal prescription fixes the homogeneous response data of the original nonlocal theory, that route did not provide the clean nonlinear two-degree-of-freedom Hamiltonian closure sought here.

The final architecture therefore removes the Φ, λ sector from the fundamental theory and replaces it with

$$\boxed{\text{MOND response} \longrightarrow \text{elliptic constraint} \longrightarrow \text{second-class reduction} \longrightarrow 2 \text{ tensor DOF.}} \quad (2)$$

3 ADM variables and phase space

Write the metric in ADM form,

$$ds^2 = -N^2 c^2 dt^2 + \gamma_{ij} (dx^i + N^i dt)(dx^j + N^j dt). \quad (3)$$

The canonical gravitational variables are

$$(\gamma_{ij}, \pi^{ij}), \quad (N, \pi_N), \quad (N^i, \pi_i). \quad (4)$$

There are therefore

$$6 + 6 + 1 + 1 + 3 + 3 = 20 \quad (5)$$

phase-space dimensions per spatial point.

The construction retains spatial diffeomorphism covariance but does not retain GR refoliation invariance. This is a preferred-foliation Hamiltonian MMG theory.

4 Scalar metric variable

Define

$$q = \frac{1}{6} \ln \det \gamma_{ij}, \quad p = \frac{\pi}{\sqrt{\gamma}}, \quad \pi = \gamma_{ij} \pi^{ij}. \quad (6)$$

The canonical normalization used in the constraint analysis is

$$\{q(\mathbf{x}), p(\mathbf{y})\} = \frac{1}{2\sqrt{\gamma}} \delta^3(\mathbf{x} - \mathbf{y}). \quad (7)$$

For an inhomogeneous Fourier mode,

$$\{D^2 q(\mathbf{k}), D^2 p(-\mathbf{k})\} = \frac{1}{2} k^4, \quad (8)$$

so the scalar pair is nondegenerate for $k \neq 0$.

5 MOND constitutive function

The target interpolation is

$$\boxed{\mu(y) = 1 - e^{-y}}. \quad (9)$$

It has the asymptotic limits

$$\mu(y) = y + O(y^2) \quad (y \ll 1), \quad \mu(y) \rightarrow 1 \quad (y \gg 1). \quad (10)$$

A primitive convenient for the Hamiltonian construction is

$$\boxed{\mathcal{G}(y) = y^2 + 2(1 + y)e^{-y} - 2}. \quad (11)$$

Then

$$\mathcal{G}'(y) = 2y - 2ye^{-y}, \quad (12)$$

$$\frac{\mathcal{G}'(y)}{2y} = 1 - e^{-y} = \mu(y). \quad (13)$$

Moreover,

$$\mathcal{G}''(y) = 2(y + e^y - 1)e^{-y} > 0 \quad (y > 0). \quad (14)$$

6 Acceleration variable and MOND constraint

Define the dimensionless acceleration variable from the lapse,

$$\boxed{y = \frac{c^2}{a_0} \sqrt{D_i \ln N D^i \ln N}}. \quad (15)$$

In the weak-field expansion

$$N = 1 + \frac{\Psi}{c^2} + O(c^{-4}), \quad (16)$$

we have

$$c^2 D_i \ln N = D_i \Psi + O(c^{-2}), \quad (17)$$

and hence

$$y = \frac{|\nabla\Psi|}{a_0} + O(c^{-2}). \quad (18)$$

The defining MOND constraint is

$$\mathcal{C}_M = \frac{c^4}{4\pi G} \sqrt{\gamma} D_i [\mu(y) c^2 D^i \ln N] - \sqrt{\gamma} \rho_m c^2 \approx 0. \quad (19)$$

Thus the lapse is determined elliptically rather than evolving as an independent scalar mode.

7 Exact Newtonian limit

Substituting the weak-field lapse gives

$$D_i \ln N = \frac{D_i \Psi}{c^2} + O(c^{-4}), \quad (20)$$

so the explicit factors of c^2 cancel and the constraint becomes

$$\nabla \cdot \left[(1 - e^{-|\nabla\Psi|/a_0}) \nabla\Psi \right] = 4\pi G \rho_b. \quad (21)$$

This reproduces the intended MOND modified Poisson equation exactly.

For $|\nabla\Psi| \ll a_0$,

$$\nabla \cdot \left[\frac{|\nabla\Psi|}{a_0} \nabla\Psi \right] \simeq 4\pi G \rho_b, \quad (22)$$

while for $|\nabla\Psi| \gg a_0$,

$$\nabla^2 \Psi \simeq 4\pi G \rho_b. \quad (23)$$

8 Ellipticity of the MOND operator

Let

$$A^{ij} = \mu(y) \gamma^{ij} + y \mu'(y) \hat{u}^i \hat{u}^j, \quad (24)$$

where $\hat{u}_i$ is the unit vector along $D_i \ln N$. The principal eigenvalues are

$$\lambda_\perp = \mu(y) = 1 - e^{-y}, \quad (25)$$

$$\lambda_\parallel = \mu(y) + y \mu'(y) = 1 - e^{-y} + y e^{-y}. \quad (26)$$

For $y > 0$,

$$\lambda_\perp > 0, \quad \lambda_\parallel > 0, \quad (27)$$

so the lapse equation is elliptic on the generic branch. At exactly $y = 0$ both eigenvalues vanish and the operator degenerates. This branch is therefore not included in the generic-branch theorem.

9 Second-class constraints

The scalar constraint set is represented by

$$\boxed{\mathcal{S}_4 = \pi_N, \quad \mathcal{S}_1 = \mathcal{C}_M, \quad \mathcal{S}_2 = D^2 q, \quad \mathcal{S}_3 = D^2 p.} \quad (28)$$

The four-by-four Dirac matrix has the relevant principal structure

$$\Delta = \begin{pmatrix} 0 & L_N & 0 & 0 \\ -L_N & 0 & b & c \\ 0 & -b & 0 & K \\ 0 & -c & -K & 0 \end{pmatrix}, \quad (29)$$

where L_N is the lapse principal operator and K is the scalar-pair bracket.

Its Pfaffian is

$$\text{Pf } \Delta = L_N K, \quad (30)$$

so

$$\boxed{\det \Delta = (L_N K)^2.} \quad (31)$$

The cross-couplings b and c do not affect the determinant.

Therefore the scalar constraint sector has rank four whenever

$$L_N \neq 0, \quad K \neq 0. \quad (32)$$

The explicitly checked generic branch is

$$\boxed{y > 0, \quad k \neq 0.} \quad (33)$$

10 Degree-of-freedom count

The 20-dimensional ADM gravitational phase space is reduced by the spatial-diffeomorphism constraint structure and the four scalar second-class constraints. The count is

$$20 - 12 - 4 = 4 \quad (34)$$

remaining phase-space dimensions, giving

$$\boxed{N_{\text{grav}} = 2.} \quad (35)$$

The surviving local variables are the transverse-traceless tensor pair. The inhomogeneous scalar pair and the lapse pair are removed by second-class constraints.

The homogeneous $k = 0$ scalar modes are not removed by $D^2 q$ and $D^2 p$ and therefore constitute a separate cosmological-background sector. They are not included in the generic-branch theorem.

11 Constraint preservation

The preservation equations are

$$\dot{\mathcal{S}}_A = \{\mathcal{S}_A, H_T\} \approx 0. \quad (36)$$

They take the linear algebraic form

$$\Delta_{AB} \lambda^B + r_A = 0, \quad (37)$$

where $r_A = \{S_A, H_{\text{can}}\}$. Since $\det \Delta \neq 0$ on the generic branch, the four multipliers are uniquely determined.

The Dirac algorithm therefore terminates without a tertiary constraint on the generic branch. This is the decisive Hamiltonian closure calculation: the constraint set is preserved without generating a new independent condition.

12 Tensor sector

The transverse-traceless perturbation satisfies

$$\partial_i h_{ij}^{TT} = 0, \quad h_{ii}^{TT} = 0. \quad (38)$$

At quadratic order the MOND constraint does not modify the TT kinetic operator. The tensor action retains the Einstein form,

$$S_{TT}^{(2)} = \frac{c^3}{64\pi G} \int d^4x \left[\frac{1}{c^2} \dot{h}_{ij}^{TT} \dot{h}_{ij}^{TT} - \partial_k h_{ij}^{TT} \partial_k h_{ij}^{TT} \right]. \quad (39)$$

Therefore

$$\boxed{\omega^2 = c^2 k^2, \quad c_T = c, \quad Q_T > 0.} \quad (40)$$

The generic linearized gravitational spectrum consequently contains the two usual tensor polarizations and no propagating gravitational scalar in the checked branch.

13 Matter coupling

A spatially covariant matter Hamiltonian can be written schematically as

$$H_m = \int d^3x (N\rho_m + N^i j_i). \quad (41)$$

For a canonical scalar,

$$\mathcal{H}_m = \frac{\pi_\chi^2}{2\sqrt{\gamma}} + \sqrt{\gamma} \left[\frac{1}{2} \gamma^{ij} \partial_i \chi \partial_j \chi + V(\chi) \right], \quad (42)$$

with positive momentum Hessian

$$\frac{\partial^2 \mathcal{H}_m}{\partial \pi_\chi^2} = \frac{1}{\sqrt{\gamma}} > 0. \quad (43)$$

The slow-dust limit reproduces the ordinary Newtonian source ρ_b and the standard continuity equation.

However, the model is not a standard four-dimensional generally covariant matter theory. Because the lapse equation is the MOND constraint rather than the GR Hamiltonian constraint, the usual Bianchi identity does not automatically yield $\nabla_\mu T^{\mu\nu} = 0$ as an off-shell identity. The relativistic matter equations therefore acquire a MOND-induced correction. This is an open phenomenological issue rather than a contradiction in the Newtonian regime.

14 Relation to minimally modified gravity

The construction is a Hamiltonian MMG theory with a preferred foliation. It should not be advertised as a manifestly four-dimensional generally covariant local metric theory. This is

consistent with the defining architecture of MMG, whose purpose is to alter the constraint structure while retaining spatial covariance and only the two tensorial gravitational degrees of freedom. [1, 2]

The four-auxiliary-constraint formalism is particularly relevant because the Hamiltonian analysis shows that four scalar auxiliary constraints can produce a two-degree-of-freedom theory without the additional minimalizing or symmetrizing condition required in some lower-constraint constructions. [2]

The 2026 Laplacian-constraint formulation further shows how spatial Laplacians of auxiliary multipliers can remove homogeneous ambiguity while maintaining the inhomogeneous scalar constraint structure. [3]

15 Falsification audit

The candidate was subjected to a reproducible twelve-gate symbolic audit:

1. constitutive primitive;
2. Newtonian limit;
3. Dirac matrix structure;
4. ellipticity;
5. Laplacian scalar-pair bracket;
6. full generic-branch rank;
7. degree-of-freedom count;
8. constraint preservation;
9. tensor sector;
10. matter consistency;
11. action/Legendre reconstruction;
12. explicit falsification criteria.

All implemented gates returned PASS on the generic branch. The audit was executed from the repository's `openai_push/final_closure` workspace with exit code 0 and the final summary **ALL GATES PASS**. The repository records the result conservatively as **CONDITIONALLY CLOSED** rather than **CLOSED**.

16 Scope of the closure claim

The correct theorem-like statement is restricted to the nondegenerate generic branch,

$$\boxed{y > 0, \quad k \neq 0.} \tag{44}$$

On that branch, the checked Hamiltonian constraint structure yields two local gravitational tensor degrees of freedom and the MOND constraint reproduces the exact interpolation function.

The following remain open:

1. a controlled $k = 0$ homogeneous/cosmological prescription;
2. a separate treatment of the exactly degenerate $y = 0$ branch;
3. a decision on the physical interpretation and acceptability of the preferred foliation;
4. restoration or characterization of the relativistic matter-conservation defect;
5. global nonlinear and cosmological viability, including structure formation, lensing, post-Newtonian tests, and strong-field behavior.

17 Conclusion

A constraint-first MOND modification of gravity has been constructed within the Hamiltonian MMG framework. Its defining phenomenological input is

$$\boxed{\mu(y) = 1 - e^{-y}}, \quad (45)$$

and its defining dynamical mechanism is the replacement of the usual lapse/Hamiltonian constraint structure by a nonlinear elliptic MOND constraint together with a second-class scalar pair.

On the generic branch, the Dirac matrix is nondegenerate,

$$\det \Delta = (L_N K)^2 \neq 0, \quad (46)$$

constraint preservation fixes the relevant multipliers without generating a tertiary constraint, and the phase-space count is

$$20 - 12 - 4 = 4, \quad (47)$$

corresponding to

$$\boxed{N_{\text{grav}} = 2.} \quad (48)$$

The tensor sector retains

$$\boxed{c_T = c, \quad Q_T > 0.} \quad (49)$$

The Newtonian limit is exactly

$$\boxed{\nabla \cdot \left[\left(1 - e^{-|\nabla \Psi|/a_0} \right) \nabla \Psi \right] = 4\pi G \rho_b.} \quad (50)$$

The appropriate scientific status is therefore

$$\boxed{\text{CONDITIONALLY CLOSED.}} \quad (51)$$

The construction constitutes a definite field-theory candidate with an explicitly checked generic two-degree-of-freedom branch and an exact MOND weak-field constitutive law. It should not yet be presented as a globally closed, manifestly four-dimensional covariant replacement for general relativity.

Reproducibility and repository

The computational audit accompanying this paper is maintained at

<https://github.com/carlzimmerman/zimmerman-formula/>

with the closure workspace under `openai_push/final_closure`. The exact repository state associated with the conditional-closure result is commit `c3206f6`.

References

- [1] Z.-B. Yao, M. Oliosi, X. Gao, and S. Mukohyama, *Minimally modified gravity with an auxiliary constraint: A Hamiltonian construction*, Phys. Rev. D **103**, 024032 (2021), arXiv:2011.00805, DOI: 10.1103/PhysRevD.103.024032.
- [2] Z.-B. Yao, M. Oliosi, X. Gao, and S. Mukohyama, *Minimally modified gravity with auxiliary constraints formalism*, Phys. Rev. D **107**, 104052 (2023), arXiv:2302.02090, DOI: 10.1103/PhysRevD.107.104052.
- [3] J. Sangtawe, A. De Felice, and K. Karwan, *Minimally modified gravity with Laplacian auxiliary constraints and an inflationary realization*, arXiv:2607.26031 (2026).